

IC-42 STANDING WAVES ON A STRING

04/23/2023

FOR IN-CLASS ONLY

42-1 OBJECTIVE

To study standing waves on a string and see the effects of changing the tension in the string, frequency and wavelength.

42-2 EQUIPMENT

String Vibrator WA-9857

String

Pulley

Mass Hanger

Masses

Meter Stick

Sine Wave Generator / Interface 850

Force Sensor



42-3 THEORY

Standing waves are present in all stringed instruments, as well as in other places such as in tuning forks. In this experiment we will see how the length of the string, frequency and tension in the string are related.

Standing waves (also called stationary waves) are produced when two traveling waves of same wavelength and speed travel in opposite directions through the same medium. Figure 1 shows a taut string, where a mechanical vibrator produces a wave on a string at one end, which moves to the other end and are reflected from a fixed support that produces a returning wave. We thus have two waves traveling in opposite directions. These waves interfere with each other to produce standing waves. At locations on the string where the two waves are 180° out of phase, very little motion occurs (no motion if the amplitudes of the two waves are the same). Such places are called nodes.



Figure 1: Standing Wave Photo, showing 3 antinodes and 4 nodes (third harmonic)

At locations where the two waves are in phase, the motion is maximum. These positions are called anti-nodes.

Figure 2 shows different representations of a standing wave. They match well what the eye or camera sees since the string speed is minimum at the extremes. The node-to-node distance is one-half of the wavelength, so in the length L , there are $n\lambda/2$ waves, where ' n ' is the number of antinodes, or harmonics in the wave. For a taut string, there must be a node at each end when there is a standing wave pattern in the string. Hence the number of nodes will be one more than the number of antinodes. For this experiment, one fixed end where the string attaches to the mechanical vibrator is not exactly a node since the wand vibrates up and down a little. However, the error involved is small, and we will neglect it.

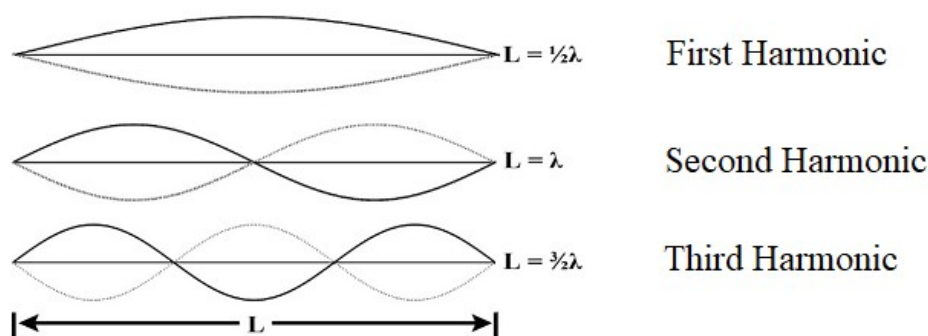


Figure 2: Modes of Vibration

For any wave with wavelength λ and frequency f , the speed, v , is

$$v = \lambda f \quad (1)$$

The speed of a wave on a string is also related to the tension in the string, T , and the linear density (=mass/length), μ , by

$$v^2 = T/\mu = \lambda^2 f^2 \quad (2)$$

L is the length of the string and n is the number of segments, antinodes, or harmonics. Since a segment is $1/2$ wavelength then

$$\lambda = 2L/n \quad \text{where } n = 1, 2, 3, \dots \quad (3)$$

Solving Equation 2 for the tension yields:

$$T = \mu \lambda^2 f^2 \quad (4)$$

Which can also be written as:

$$f = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}} \quad (5)$$

42-4 EQUIPMENT DESCRIPTION

String Vibrator

The String Vibrator has a small metal wand that vibrates up and down. The vibration is caused by an alternating voltage signal provided by a separate power source, which may be the Sine Wave Generator or the Pasco 850 interface.



Sine Wave Generator

The Sine Wave Generator produces an adjustable frequency range of 1-800 Hz, and an adjustable output voltage. It works with a 15V DC power adapter. The output is fed to the String Vibrator. Start with a low output voltage (about 5 V), but do not exceed 10 V.



Interface 850

Connect the Interface 850 to the computer, turn on Capstone Software. In Capstone, click the Signal Generator at the left side of the screen. Open Output 1 and set it to Sine Waveform at an Amplitude (i.e. voltage) of 5.00 V and a frequency of 60 Hz. When you are ready to start the experiment, click 'On' to turn power on. Increasing the voltage will increase the amplitude of vibration of the wand in the String Vibrator. You may try different values to see what works best for you, but do not exceed 10 V. If the String Vibrator begins to clatter, you should decrease the Amplitude.



Force Sensor

The Force Sensor measures the force with which it is being pulled, i.e. the Tension in the string. You have to connect it to the Interface, and when there is no force on it, press the Tare button to set the force at zero. In Capstone you can make a Table to see the value of the force.



String Linear Mass Density

Since the string is very light, we can measure the mass of a long piece and then cut it into the required lengths that will have nearly the same mass density as average mass density of the long piece (assuming that the string is of uniform thickness).

1. Cut a piece of string that is long enough to provide each group with about 1.2 m of string. That is, if there are five lab groups, cut a piece of string that is about 6 m long (or longer).
2. Use the tape measure to determine the length of the string to a precision of ± 1 cm.

3. Use the best available scale to determine the mass of the string. Record the mass and calculate the mass/length.
4. The uncertainty in the “mass density” arises from the measurement of the “String mass”. If the “String mass” was 2.0 g measured with a single pan balance, the precision is 0.1 g and the uncertainty would be $(0.1 \text{ g}/2.0 \text{ g}) \times 100 = 5\%$. Similarly, using a good digital scale, the numbers might be $2.00 \text{ g} \pm 0.01 \text{ g}$ and the uncertainty would be $(0.01 \text{ g}/2.00 \text{ g}) \times 100 = 0.5\%$. However, you should not claim an uncertainty less than 1% since the string can vary by 1% in its mass density. Record your estimate for the uncertainty.
5. Cut a piece of string about 1.2 m long for each lab group.

42-5 PROCEDURE

CASE A: Constant Tension, Changing Frequency

1. Setup the equipment as shown in Figure 3. Clamp the String Vibrator and pulley to the table. Connect the two wires from the string generator to Output-1 on the 850 Pasco interface (or to the Sine Wave Generator).
2. Tie the string to the metal wand of the String Vibrator. Keep the distance between the knot on the string vibrator and the top of the pulley to about 1.00 meter.
3. Pass the string over the pulley and attach a mass of 200 grams to create the tension. Keep this mass fixed in this part of the experiment.
4. Now click on the Signal Generator at the left of the Pasco Capstone screen. Obtain a 5.0 V sine wave in output 1. Click ‘ON’ to start supplying power to the string vibrator.
5. Now adjust the frequency in the Signal Generator until you get a standing wave in one segment (i.e. the first harmonic). Note this frequency, and wavelength.
6. Increase the frequency gradually until you obtain a standing wave in the 2nd, 3rd, 4th, and 5th harmonic. Record each wavelength and measured frequency.
7. Calculate the frequency from equation 5. Find the percent difference in the measured and calculated frequencies. Discuss in your report the possible causes of any error that you have.
8. Make a plot of measured frequency on the y-axis versus $1/\text{wavelength}$ on the x-axis. This should be a straight line whose slope is the speed of the waves. Now calculate the value of $\sqrt{T/\mu}$, which is also the speed of the waves in the string. Calculate the percent difference in the two speeds.

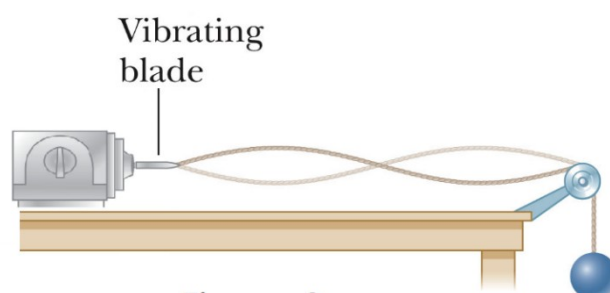


Figure 3

CASE B: Constant Frequency, Changing Tension:

1. Setup the equipment as shown in Figure 4. Clamp the String Vibrator to the table. Tie the string to the string Vibrator and the other end to the Force Sensor. Connect the Force Sensor to the Interface 850 and open a table to see the value of the force. Keep the distance between the knot on the string vibrator and the knot on the force sensor to about 1.00 meter.
2. Set up a sine wave of Amplitude of 5.00 V and a frequency of 60 Hz. Click 'ON' to start supplying power to the string vibrator. This frequency is to remain constant in this part of the experiment.
3. Now hold the force sensor and gently pull the string. You would see in Capstone Software that the tension increases. Adjust the tension until you get a standing wave in one segment (i.e. the first harmonic). Note this tension and wavelength.
4. Increase the tension gradually until you obtain a standing wave in the 2nd, 3rd, 4th, and 5th harmonic. Record each tension and wavelength.
5. Calculate the tension from equation 4. Compare with the measured values of Tension and find the percent difference. Discuss in your report the possible causes of any error that you have.
6. Make a plot of measured Tension on the y-axis versus λ^2 on the x-axis. This should be a straight line with a slope of μf^2 : $T = \mu \lambda^2 f^2$ hence $T = (\mu f^2) \lambda^2$.
7. **Alternate to using the Force Sensor:** Steps 1 to 4 can be done by using the setup as in Case A, keep the frequency constant at 60 Hz, and slowly add weights to the weight hanger to increase the tension in the string. Note the weights (= tension) for the first to the fifth harmonic.



Figure 4

CASE C: Constant Wavelength (Second Harmonic), Changing Frequency and Tension:

1. Use the same setup of the equipment as shown in Figure 4.
2. Set up a sine wave of Amplitude of 5.00 V and a frequency of 60 Hz. Click 'ON' to start supplying power to the string vibrator.
3. Now hold the force sensor and gently pull the string. Adjust the tension until you get a standing wave in two segments (i.e. the second harmonic). Note this tension and wavelength.
4. Now change the frequency in steps of 30 Hz, adjust the tension by pulling the force sensor, until you again obtain a standing wave in the second harmonic. Record each frequency and tension.

5. Calculate the tension from equation 4. Compare with the experimental values and find the percent difference. Discuss in your report the possible causes of any error that you have.
9. Make a plot of Measured Tension on the y-axis versus f^2 on the x-axis. This should be a straight line with a slope of $\mu\lambda^2$: $T = \mu\lambda^2 f^2$ or $T = (\mu\lambda^2) f^2$
10. **Alternate to using the Force Sensor:** Steps 1 to 4 can be done by using the setup as in Case A. Set the frequency to 60 Hz, and slowly add weights to the weight hanger to increase the tension in the string until you get standing waves in the second harmonic. Note the values. Now increase the frequency in steps of 30 Hz and for each frequency, add weights to increase the tension until you again get standing waves in the second harmonic. Record the values at each step.

42-6 PRECAUTIONS

1. The amplitude of vibration of the vibrating blade should be small, so keep the voltage to the vibrator less than 10 V.
2. As you increase tension in the string it may stretch if it is elastic. This will reduce the mass/length (μ). See how you can make an appropriate correction for this.
3. Make sure that the hanging weights are not swinging, as that will change the tension.
4. If the String Vibrator is placed horizontally on the table, its blade vibrates in the vertical direction, the standing waves are not easily seen if you are looking from directly above the string. See if you can place the String Vibrator in the vertical position, so that the wand vibrates in the horizontal direction.
5. If the string is white or light colored, a black or dark background is better for seeing the standing waves.
6. If you are using the force sensor to pull the string, keep your hand steady.

42-7 DATA TABLE

Length of the string: _____ Mass of the string: _____
 Calculated mass / length of string = μ = _____ Mass of weight hanger (M_h): _____

Case A: Constant Tension, Changing Frequency and Harmonics

Mass added to weight hanger = M_a = _____ Total hanging mass = $M = M_a + M_h$ = _____

Tension in the string = $T = Mg$ = _____ Length of vibrating string = L = _____

Number of Harmonic (n)	Number of nodes ($=n+1$)	Wavelength $\lambda = 2L/n$ (m)	Measured Frequency f (Hz)	Calculated Speed of wave $V = \lambda * f$ (m/s)	Calculated Frequency $f = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$ (Hz)	Percent difference in frequency
1						
2						
3						
4						
5						

Measured value of speed = Slope of measured f vs $1/\lambda$ graph = _____ (attach graph)

Calculated Speed = $\sqrt{T/\mu}$ = _____ % difference in the two values of speed = _____

Case B: Constant Frequency, Changing Tension and Harmonics

Length of vibrating string = L = _____ m Frequency (constant) = f = _____ Hz

Number of Harmonic (n)	Number of nodes ($=n+1$)	Wavelength $\lambda = 2L/n$ (m)	Measured Tension in the String T (N)	Calculated Speed of wave $V = \lambda * f$ (m/s)	Calculated Tension $T = \mu \lambda^2 f^2$ (N)	Percent difference in Tension
1						
2						
3						
4						
5						

Measured value of μf^2 = Slope of Measured T versus λ^2 = _____ (attach Graph)

Calculated value of $\mu f^2 =$ _____ % difference in the two values of $\mu f^2 =$ _____

Case C: Constant Wavelength (Harmonics), Changing Frequency and Tension.

Length of vibrating string = $L =$ _____ m Harmonic = $n =$ __

Wavelength = $\lambda = 2L/n =$ _____ m

Frequency f (Hz)	Measured Tension in the String T (N)	Calculated Speed of wave $V = \lambda * f$ (m/s)	Calculated Tension $T = \mu \lambda^2 f^2$ (N)	Percent difference in Tension

$\mu \lambda^2 =$ Slope of Measured T versus f^2 graph= _____ (Attach Graph)

Calculated value of $\mu \lambda^2 =$ _____ % difference in the two values of $\mu \lambda^2 =$ _____

42-8 REPORT SUBMISSION

Upload the following in the Report for this Lab:

		Points in report
1.	The completely filled up "Report Form". (Includes accuracy of measurements, units, significant figures, performance / effort).	$3 \times 10 = 30$
2.	Graphs for the three cases showing the best fit line. Don't forget to label the axes. The data on the graphs should be legible.	$3 \times 5 = 15$
3.	Sample Calculations.	5
4.	Sources of Error in this experiment. Indicate the major sources of error. Make a List.	5
5.	Discussion of Results	10
	Total	65

Extra Credit: Attach a video of experiment to get up to 5 points EC.

42-9 ADDITIONAL INFORMATION

PhET Simulation Wave on a String:

<https://phet.colorado.edu/en/simulation/wave-on-a-string>

Standing waves in a spring

0:40 min http://www.youtube.com/watch?v=3BN5-JSsu_4&feature=fww

standing waves (string on floor – teacher with students))

3:48 min http://www.youtube.com/watch?v=603qpamkV_Y

Standing waves in water (wave tank)

4:42 min <http://www.youtube.com/watch?v=NpEevfOU4Z8>

42-10 POINTS TO THINK ABOUT

1. Waves transfer energy. Does a standing wave also transfer energy? If so, in which direction?
2. When you pluck a guitar string, do standing waves set up in it? Is the frequency of the sound coming from the guitar string the same as the frequency of standing waves in it?
3. Can a string have standing waves of more than one harmonic on it at the same time?

4. Why are the amplitudes of the standing waves larger than that of the vibrating wand?
- 5.

42-11 SAMPLE DATA

DATA TABLE

(Data from Jonathon, Angel, Cynthia and Edwin, CRN 11937)

Length of the string: 1.30 [m]

Mass of the string: 27 [g]

Calculated mass / length of string = $\mu = 0.0208$ kg/m

Mass of weight hanger (Mh): 4.94 [g]

Case A: Constant Tension, Changing Frequency and Harmonics

Mass added to weight hanger = $M_a = 200$ [g]

Total hanging mass = $M = M_a + M_h = 204.94$ [g]

Tension in the string = $T = M g = 2.01$ [N]

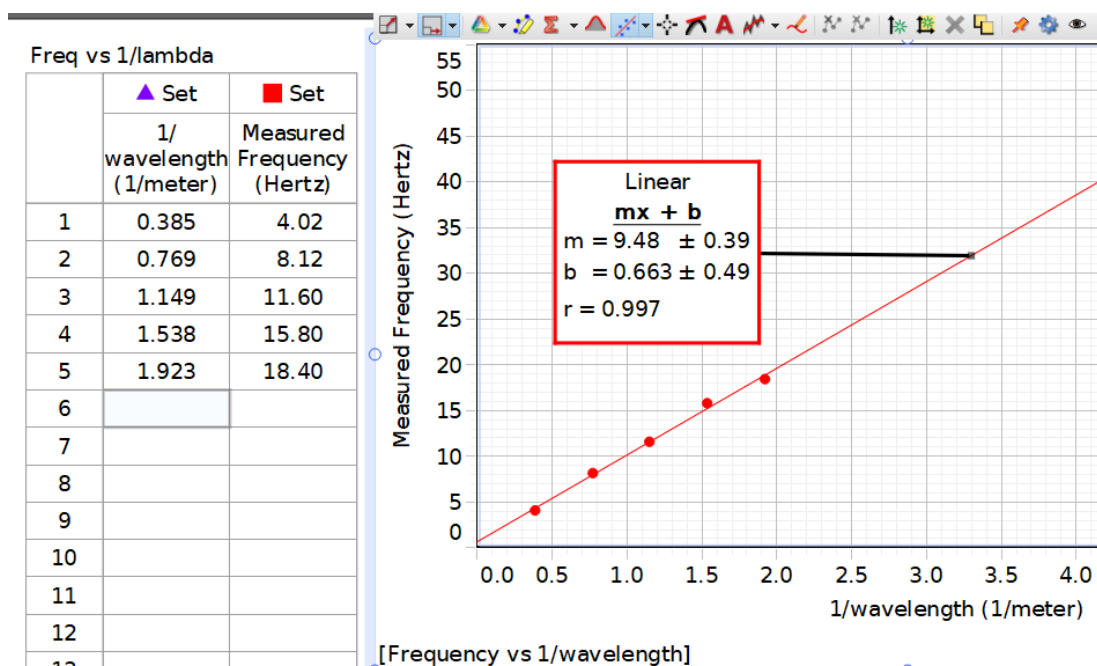
Length of vibrating string = $L = 1.3$ [m]

Number of Harmonic (n)	Number of nodes	Wavelength $\lambda = 2L/n$ (m)	Measured Frequency f (Hz)	Speed of wave $V = \lambda * f$ (m/s)	Calculated Frequency $f = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$ (Hz)	Percent difference in frequency
1	2	2.6	4.02	10.45	3.78	6.2 %
2	3	1.3	8.12	10.56	7.56	7.1 %
3	4	0.87	11.6	10.09	11.3	2.6 %
4	5	0.65	15.8	10.27	15.1	4.5 %
5	6	0.52	18.4	9.57	18.9	2.7 %

$\sqrt{T/\mu} = 9.83$

Slope of Measured f vs $1/\lambda$ graph = 9.48

% difference = 3.6 %



Case B: Constant Frequency, Changing Tension and Harmonics

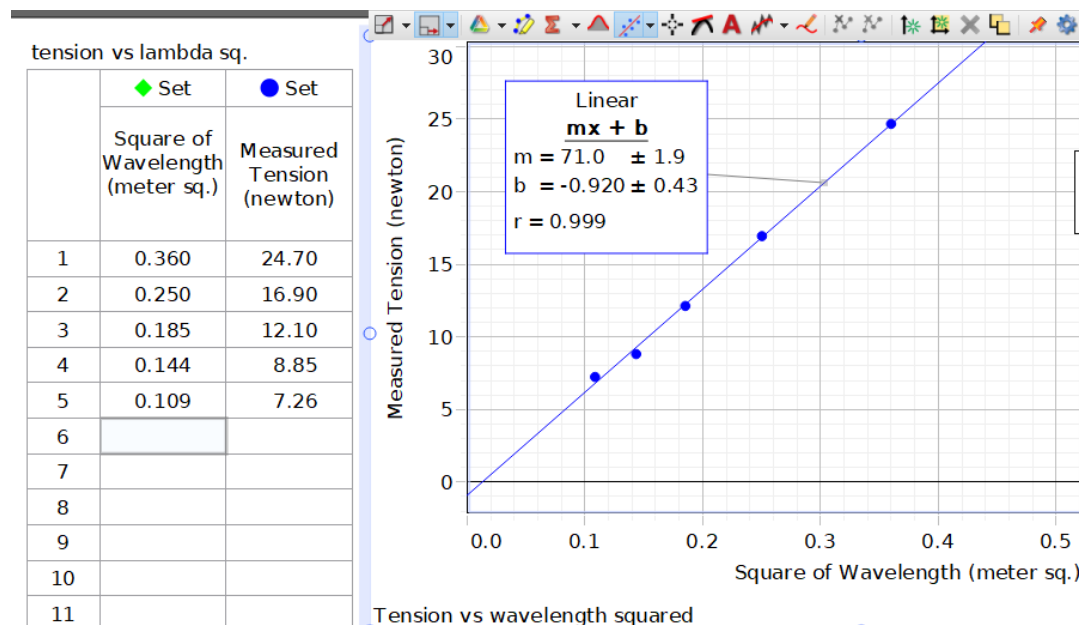
Length of vibrating string = $L = 1.50$ m

Frequency (constant) = 60 Hz

Number of Harmonic (n)	Number of nodes	Wavelength $\lambda = 2L/n$ (m)	Measured Tension in the String T (N)	Calculated Speed of wave $V = \lambda * f$ (m/s)	Calculated Tension $T = \mu \lambda^2 f^2$ (N)	Percent error in Tension
5	6	0.60	24.7	36.0	23.3	5.8 %
6	7	0.50	16.9	30.0	16.2	4.2 %
7	8	0.43	12.1	25.8	12.0	0.8 %
8	9	0.38	8.85	22.8	9.36	5.6 %
9	10	0.33	7.26	19.8	7.06	2.8 %

Calculated value of $\mu f^2 = 74.9$ N Value of $\mu f^2 = \text{Slope of Measured T versus } \lambda^2 = 71.0$ N

% difference in the two values of $\mu f^2 = 5.3$ %



Case C: Constant Wavelength (Second Harmonic), Changing Frequency and Tension:

Length of vibrating string = $L = 1.1$ m

Harmonic = 7

Wavelength = $\lambda = 2L/n = 0.314$ m

Frequency f (Hz)	Measured Tension in the String T (N)	Calculated Speed of wave $V = \lambda * f$ (m/s)	Calculated Tension $T = \mu \lambda^2 f^2$ (N)	Percent error in Tension
55	7.61	17.3	7.31	4.0 %
60	8.18	18.8	8.70	6.2 %
65	9.85	20.4	10.2	3.5 %
70	11.3	22.0	11.8	4.3 %
75	13.3	23.6	13.6	2.2 %

Calculated value of $\mu \lambda^2 = 0.00205$ kg.m

$\mu \lambda^2$ = Slope of measured T versus f^2 from graph= 0.00224 kg.m

% difference in the two values of $\mu \lambda^2 = 8.9$ %

